



DIRECTORATE OF PUBLIC INSTRUCTIONS (COLLEGES) PUNJAB
Intermediate Classes December Test - 2025

Subject: Physics

Class: 1st Year

Section: _____

Paper Type: Objective

Code: 001

Name: _____

Total Marks: 17

Time: 20 Minutes

Roll No.: _____

Q#1: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question. Cutting or filling two or more circles will result in zero marks in that question.

(1 × 17 = 17)

Sr#	Questions	Bubble
1	A precise measurement is that which have less a) Fractional uncertainty b) Percentage uncertainty c) Absolute uncertainty d) Significant figures	☐ A ☐ B ☐ C ☐ D
2	Which of the following is the most precise instrument? a) Meter rod b) Screw gauge c) Vernier calipers d) All of these	☐ A ☐ B ☐ C ☐ D
3	Which of the following are dimensionally similar? a) Angle and Strain b) Pressure and stress c) torque and energy d) All are true	☐ A ☐ B ☐ C ☐ D
4	Projection of $\vec{B}$ on $\vec{A}$ is a) $A \cos \theta$ b) $B \cos \theta$ c) $\frac{\vec{A} \cdot \vec{B}}{A}$ d) Both (b) and (c)	☐ A ☐ B ☐ C ☐ D
5	The cross product of two vectors is a negative vector when they are at an angle of a) 0° b) 180° c) 90° d) 270°	☐ A ☐ B ☐ C ☐ D
6	Which of the following quantities remains unchanged during projectile motion? a) Vertical component of acceleration b) Horizontal component of acceleration c) Horizontal component of velocity d) All of these	☐ A ☐ B ☐ C ☐ D
7	The angle of projection for which maximum height is equal to the horizontal range at angle of projection a) 15° b) 25° c) 45° d) 72°	☐ A ☐ B ☐ C ☐ D
8	If linear velocity and radius are both made half of a body moving in a circle, then centripetal force F_c becomes a) F_c b) $\frac{F_c}{2}$ c) $\frac{F_c}{4}$ d) $2 F_c$	☐ A ☐ B ☐ C ☐ D
9	The low flying satellites complete one rotation around the Earth is a) 5060 seconds b) 84 minutes c) 24 hours d) Both (a) and (b)	☐ A ☐ B ☐ C ☐ D
10	The acceleration of freely falling body is a) $+9.8 \text{ m.s}^{-2}$ b) -9.8 m.s^{-2} c) 0 m.s^{-2} d) $+19.6 \text{ m.s}^{-2}$	☐ A ☐ B ☐ C ☐ D
11	The angular displacement for daily rotation of the Earth is a) 0 rad b) $\pi \text{ rad}$ c) $2\pi \text{ rad}$ d) $4\pi \text{ rad}$	☐ A ☐ B ☐ C ☐ D
12	Conservative force from the following is	☐ A ☐ B ☐ C ☐ D



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	a) Air resistance b) Tension in the string	c) Frictional force d) Elastic spring force	
13	The dimensions of power are a) $[M L^{-1} T^{-1}]$ b) $[M L T^{-2}]$	c) $[M L^2 T^{-3}]$ d) $[M L^2 T^{-1}]$	(A) (B) (C) (D)
14	Which one is bigger unit of energy? a) <i>joule</i> b) <i>kilowatt hour</i>	c) N-m d) <i>electron – volt</i>	(A) (B) (C) (D)
15	When the opposite faces of a rigid cube are subjected to shear stress, the shear strain γ produced is given by a) $\gamma = \tan \theta$ b) $\gamma = \sin \theta$	c) $\gamma = \sec \theta$ d) $\gamma = \cos \theta$	(A) (B) (C) (D)
16	When a ship is placed in water, which force acts in accordance with the Archimedes principle to keep it floating? a) Gravitational force b) Magnetic force	c) Buoyant force d) Frictional force	(A) (B) (C) (D)
17	The strain energy can be obtained by the area under a a) Velocity – time graph b) Force – extension graph	c) Force – time graph d) Stress- strain graph	(A) (B) (C) (D)



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Total Marks: 68

Time: 2 Hours & 40 Mins

Roll No.: _____

Section-I

Q # 2: Give Short Answers to the following questions. (ANY Eight) (8×2=16)

- Show that Einstein's equation $E = mc^2$ is dimensionally correct.
- State principle of homogeneity.
- The mass of a metal box measured by a lever balance is 3.25 kg. Two silver coins of masses 10.01 g and 10.02 g measured by a beam balance are added to it. What is now the total mass of the box correct up to the appropriate precision?
- Describe dimensions of physical quantities.
- Find the dimensions of (a) Potential energy (b) Universal gravitational constant.
- Three students measured the length of a rod with a scale on which minimum division is 1 mm and recorded as: (i) 0.4235 m (ii) 0.42 m (iii) 0.424 m. Which record is correct and why?
- Round off each of the following numbers to 3 –significant figures and write your answer in scientific notation. (a) 0.02055 (b) 4656.5.
- Derive 3rd equation of motion.
- A car travelling at 10 m s^{-1} accelerates uniformly at 2 m s^{-2} . Calculate the velocity after 5 s.
- Show that range of projectile is maximum at an angle of 45° .
- Why does a hunter aiming a bird in a tree miss the target exactly at the bird?
- Define impulse and show how it is related to momentum?

Q # 3: Give Short Answers to the following questions. (ANY Eight) (8×2=16)

- State second law of motion in case of rotation.
- When a moving car turns around a corner to the left, in what direction do the occupants tend to fall? Explain briefly.
- What is the effect of changing the position of a diver while diving the pool?
- A ball is just supported by a string without breaking, if it is whirled in a vertical circle, it breaks, explain why?
- How is the centripetal force supplied in the following cases: (a) A satellite orbiting around the Earth (b) A car taking a turn on a level road?
- Prove that arc length is the product of radius of the circle and angular displacement.
- Distinguish between conservative and non – conservative forces.
- Derive work – energy theorem.
- Show that power is a scalar quantity.
- Define commercial units of electrical energy and derive its relation with SI – units.
- A body falls towards the Earth in air. Will its total mechanical energy be conserved during fall? Justify.
- Does wind contain kinetic energy? Explain.

Q # 4: Give Short Answers to the following questions. (ANY Six) (6×2=12)

- A body dropped from the height of H reaches the ground with a speed of $1.2\sqrt{gH}$. Calculate work done by air friction.



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- ii. A force acts on a ball moving with 14 m s^{-1} speed and brings its speed to 6 m s^{-1} . Has the force done positive or negative work? Explain your answer.
- iii. Why work done against friction is non – conservative in nature? Explain briefly.
- iv. Differentiate between streamline and turbulent flow of a fluid.
- v. How does pressure changes with depth in fluids?
- vi. How does an object float or sink according to Archimedes Principle?
- vii. Why standing near fast moving train is dangerous? Explain briefly.
- viii. Differentiate between stress, strain and Young's modulus. Write down their SI – units.
- ix. Distinguish between ductile and brittle substances.

Section-II

Note: Attempt any THREE of the following questions.

(8×3=24)

Q # 5: (a) What is an isolated system? State and explain law of conservation of momentum. (5+3=8)

(b) The diameter and length of a metal cylinder measured with the help of a vernier calipers of least count 0.01 cm are 1.22 cm and 5.35 cm respectively. Calculate its volume and uncertainty in it.

Q # 6: (a) Show that astronauts in a satellite are in the state of weightlessness. (5+3=8)

(b) If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, then prove that vectors $\vec{A}$ and $\vec{B}$ are perpendicular to each other.

Q # 7: (a) What is an elastic collision? Find the velocities of two bodies after head – on elastic collision. (5+3=8)

(b) A space – station has a radius of 20 cm and rotates at an angular velocity of 2 rad s^{-1} . What is the artificial gravity experienced by the astronauts on the space – station?

Q # 8: (a) What is conservative force? Prove that work done in a gravitational field along a closed path is zero. (5+3=8)

(b) An empty meteorological balloon weighs 80 N . It is filled 10 cubic meters of hydrogen. How much maximum contents the balloon can lift besides its own weight? The density of hydrogen is 0.09 kg m^{-3} and density of air is 1.3 kg m^{-3} .

Q # 9: (a) State and prove the Archimedes principle. (5+3=8)

(b) An object weighing 98 N is dropped from a height of 10 m . Its speed just before hitting the ground is 12 m s^{-1} . What is the frictional force acting on it?